Milestone 1 Report: Formula 1 Podium Prediction

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Team members: Mark Maged, Alya Ahmed, Mina Hossam, Aya El Askari

1. Introduction

We predict whether a Formula 1 driver will finish on the podium using only information available **after qualifying and before the race starts**. On the unseen 2022–2024 seasons, our selected model (random forest, depth 6) reaches **PR-AUC 0.709, ROC-AUC 0.939 and F1 0.683**, against a PR-AUC of 0.647 for the rule "the top three on the grid finish on the podium".

Item	Definition
Unit of analysis	One driver entering one race; the modeling table is unique on (raceId, driverId)
Target	`podium = 1` if `positionOrder <= 3`, else 0 (binary classification)
Prediction cutoff	After qualifying, before the start. Race results, points, laps, status, pit stops and post-race standings are excluded
Data	Formula 1 World Championship (1950–2024), Kaggle/Ergast: 14 relational CSV tables
Modeling period	2003–2024 (8,935 driver-race rows, 14.4 % podiums); the reasons are in Section 2

Artifacts:

- Kaggle notebook (single Run All): https://www.kaggle.com/code/markkmagedd/notebook0f92b4f689
- GitHub repository: https://github.com/markkmagedd/f1-podium-prediction

All figures and numbers in this report come from the Kaggle run of that notebook.

2. Data audit and cleaning

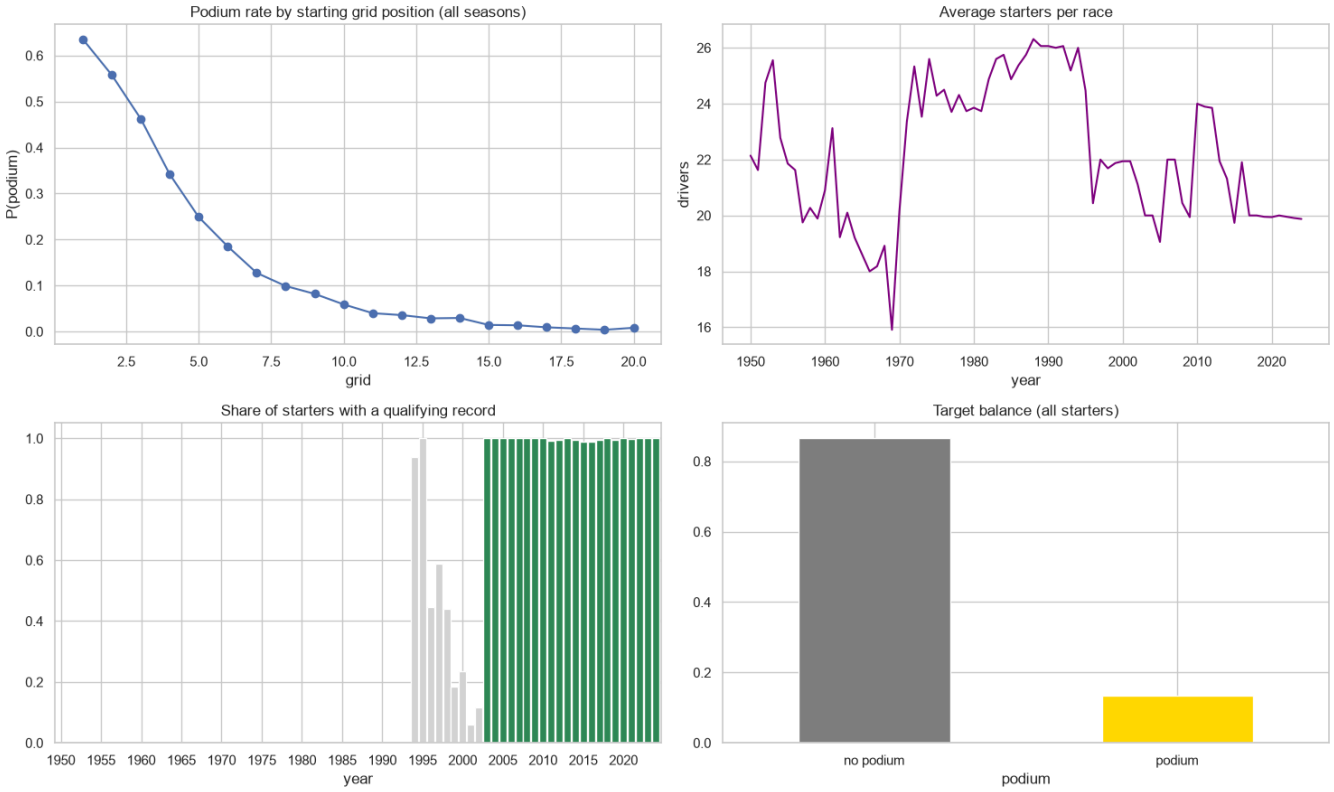
Cleaning removed every placeholder value, typed all dates and times, fixed the grain to one row per driver per race, and confirmed that all primary and foreign keys are valid. The raw CSV files were never modified; all cleaning works on copies.

Check	Before cleaning	After cleaning
Result rows	26,759	26,668
Extra duplicate (raceId, driverId) rows	91	0
Placeholder ("sentinel") strings	160,144	0
Numeric columns stored as text	73	0
Date columns typed as datetime	0	7

Sentinels and typing. Missing values appear as the literal `\N` (122,887 cells in `results` alone) and as empty strings (68 cells in `qualifying.q2/q3`). Both become `NaN`. A column is converted to a number only if no real value is lost. Dates become `datetime`. Lap, qualifying, fastest-lap and pit-stop times (`"m:ss.sss"`) become seconds; on lap times the parser matches the file's own `milliseconds` column to within 10^{-10} ms.

Grain. 85 (raceId, driverId) pairs (176 rows) are duplicated, all in 1950–1978 shared drives, where drivers swapped cars mid-race. **Rule:** keep the entry the driver started in, decided by pre-race information only: a real grid slot first, then the best grid position, then the first listed entry. An outcome-based rule ("keep the best finish") would select rows by race result, a form of leakage; it would change the podium label of 17 of the 85 pairs. An assertion guarantees uniqueness.

Keys. The primary keys of all 14 tables, including composite keys for `lap_times` and `pit_stops`, are unique and non-null. All 23 foreign-key links reconcile, with 0 orphan values.



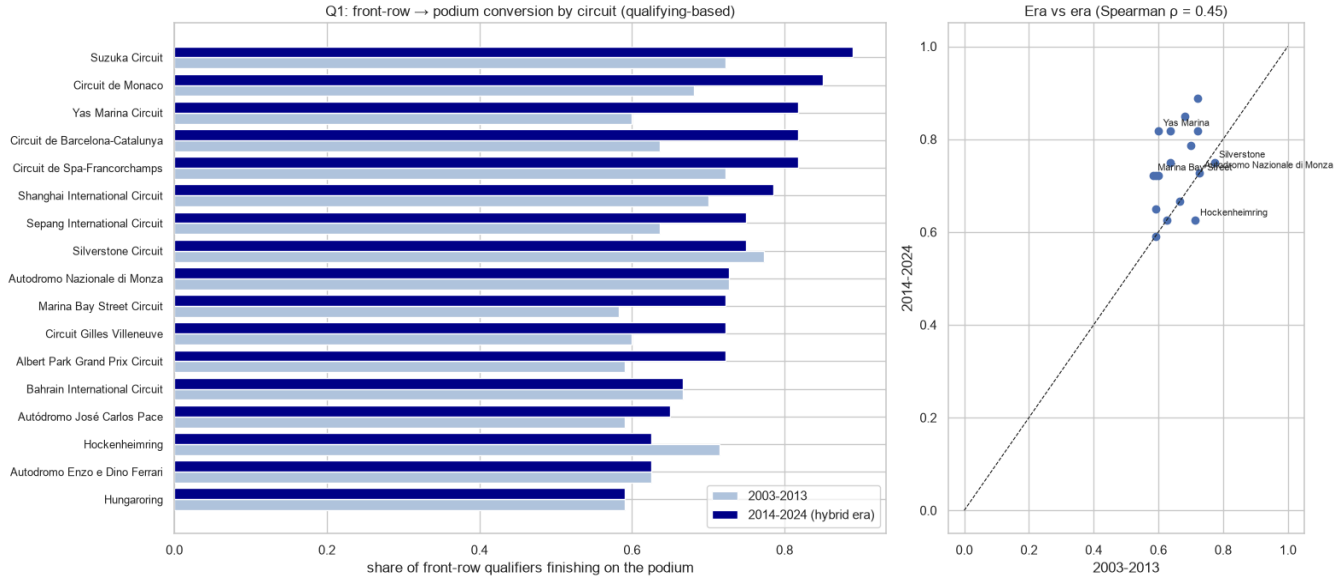
EDA findings and the decisions they drove:

- **Grid position dominates.** The podium rate falls from 64 % on pole to near 0 beyond 12th, so grid and qualifying form the core feature groups.
- **Qualifying coverage drift.** No qualifying records exist before 1994, coverage is 6–59 % in 1996–2002, and it is 100 % from 2003. **The modeling period is therefore 2003–2024**, so the qualifying features are never missing for structural reasons.
- **Field size changed** (up to 26 starters in the 1980s–90s, about 20 today), which further supports a consistent modern period.
- **Class imbalance.** About 13–14 % of starters reach the podium, so models are evaluated with PR-AUC and F1, not accuracy.
- 1,610 entries are non-starters (did not qualify, did not pre-qualify, withdrew). At the cutoff it is already known they will not race, so they are excluded.

3. Data-engineering questions

Each answer uses multi-hop joins (results → races → circuits, plus qualifying, drivers, status and constructors), and states its grain, its missingness and its relation to the pre-race cutoff.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?



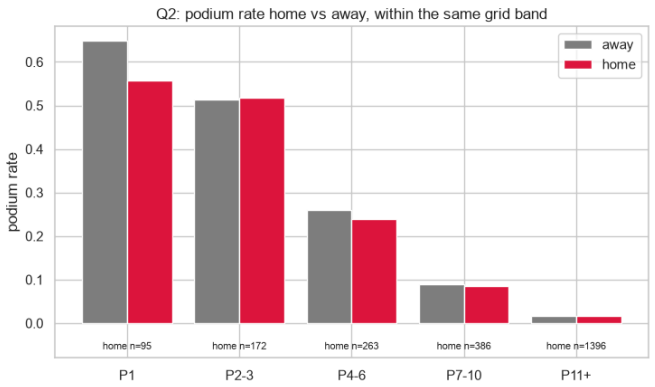
Answer. In the hybrid era (2014–2024) the front row converts best at **Suzuka (89 %)** and **Monaco (85 %)**, followed by Spa, Barcelona and Yas Marina (82 %). In 2003–2013 the leaders were **Silverstone (77 %)**, **Monza (73 %)**, and Suzuka and Spa (72 %). Overall, 70.8 % of front-row qualifiers (72.4 % of front-row starters by grid) reach the podium, and conversion rose at most circuits after 2014.

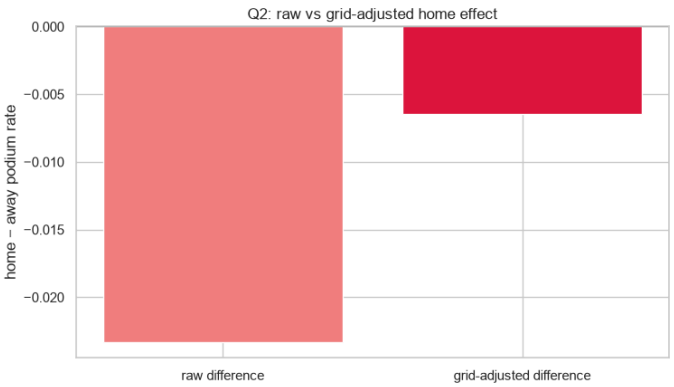
The ranking **partly changes**: the Spearman rank correlation between the eras is 0.45 ($p \approx 0.07$, 17 circuits). Silverstone fell from 1st to 7th, Yas Marina rose from 12th to 3rd, and Hockenheim fell from 5th to 15th.

Qualifying vs grid. 38 front-row qualifiers started further back after grid penalties, and only 29 % of them reached the podium. 36 drivers were promoted onto the front row by others' penalties, and 64 % of them did. Grid-based conversion therefore differs at some circuits; at Interlagos, for example, it is 79 % vs 65 % qualifying-based in 2014–2024.

Grain: one row per front-row starter (two per race). **Missingness:** qualifying is complete for 2003–2024, so both eras are equally covered; circuits with fewer than 8 front-row starts in either era are excluded. **Cutoff:** qualifying position and grid are both known before the race.

Q2. Do drivers podium more often at home, after controlling for starting position?

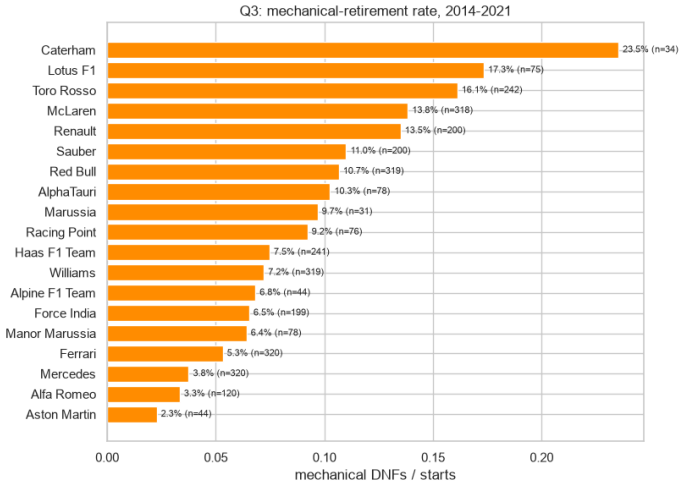


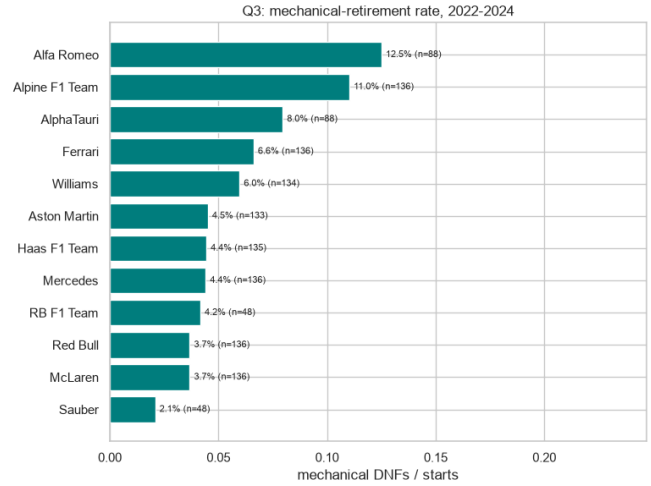


Answer: no. Raw podium rates are 11.2 % at home vs 13.6 % away; home drivers start further back on average. Within the same grid band the rates are nearly identical (P2–3: 51.7 % vs 51.3 %; P7–10: 8.5 % vs 8.9 %), and the grid-adjusted difference is −0.6 percentage points. A logistic regression `podium ~ home + grid band + era` gives an **odds ratio of 0.92 (95 % CI 0.78–1.08, p = 0.30)**. The interval includes 1, so there is no reliable home effect.

Grain: one row per starter with a grid slot, all seasons (22,602 rows, 2,312 at home), restricted to drivers from countries that have hosted a Grand Prix. **Missingness / mapping:** nationality was mapped to circuit country by hand, and mixed nationalities map to the first one. **Cutoff:** nationality, circuit and grid are pre-race.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?





Answer.

Rank	2014–2021	2022–2024
1	Caterham 23.5 % (34 starts)	Alfa Romeo 12.5 % (88 starts)

Rank	2014–2021	2022–2024
2	Lotus F1 17.3 % (75)	Alpine 11.0 % (136)
3	Toro Rosso 16.1 % (242)	AlphaTauri 8.0 % (88)
4	McLaren 13.8 % (318)	Ferrari 6.6 % (136)
5	Renault 13.5 % (200)	Williams 6.0 % (134)

The overall mechanical-retirement rate fell from **9.2 %** to **6.0 %** of starts. The new turbo-hybrid power units of 2014 were initially unreliable, while the 2022–2024 power units were mature. Caterham's rate rests on only 34 starts.

Definition: each `status` value was classified as mechanical (a car component failed, e.g. Engine, Gearbox, Power Unit, ERS, Hydraulics, Brakes), not mechanical (accidents, collisions, driver, disqualification, finished), or ambiguous (Retired with no cause, Damage, Debris, Puncture, wings). Only mechanical counts. **Grain:** one row per start, aggregated per constructor and period (≥ 30 starts). **Missingness:** ambiguous statuses are excluded, so these rates are lower bounds; rebrands count as separate constructors. **Cutoff:** a descriptive, post-race question. This is exactly why `status` is not allowed as a model feature.

4. Selected features and their validity at the pre-race cutoff

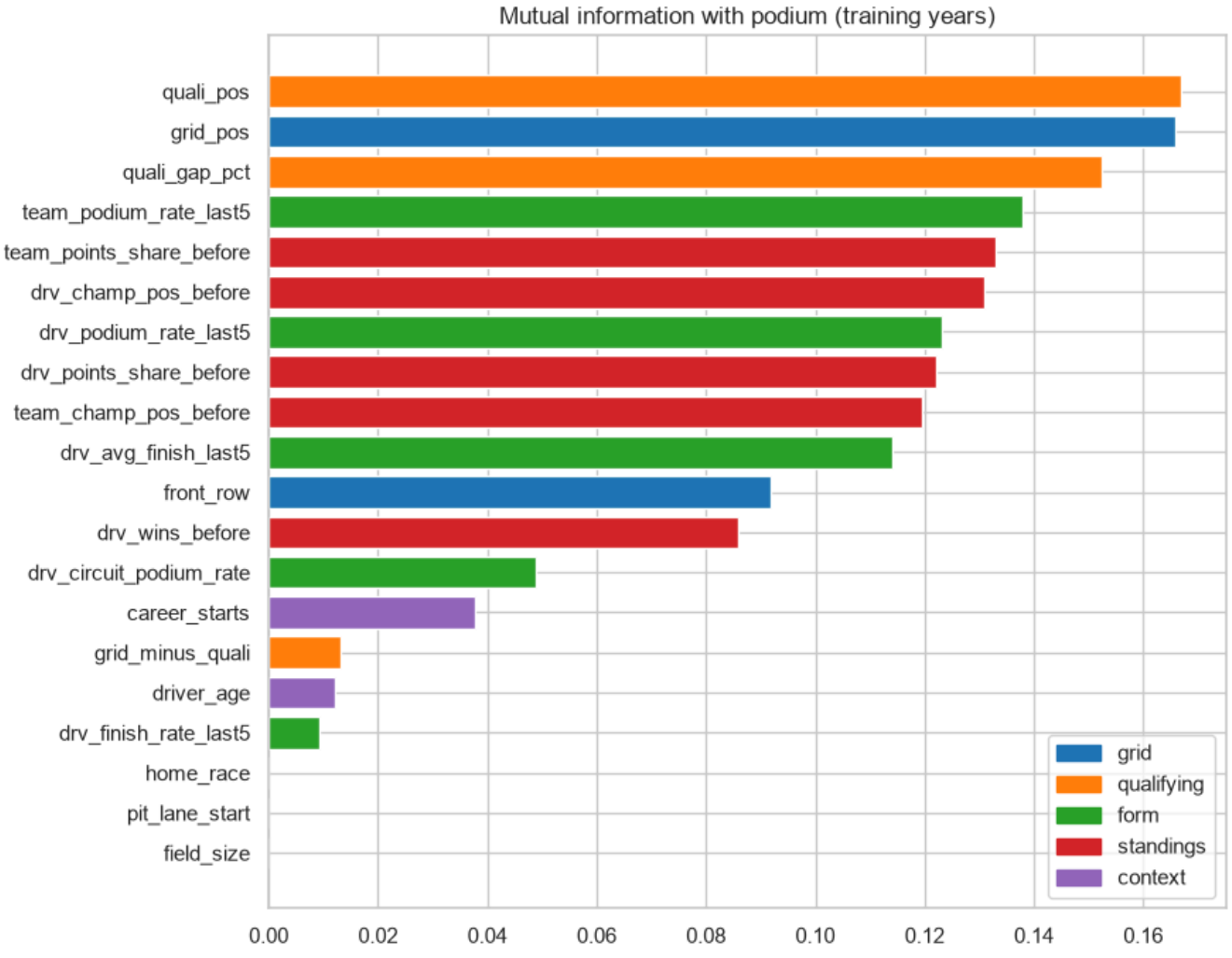
We use 20 features in five groups. Every one is known after qualifying and before the start.

Group	Feature	Definition	Why it is valid before the race
grid	`grid_pos`	Starting slot (pit-lane start = back of the field)	The grid is fixed before the start
grid	`front_row`	1 if starting 1st or 2nd	Derived from the grid
grid	`pit_lane_start`	1 if starting from the pit lane	Decided before the start
qualifying	`quali_pos`	Qualifying classification	Qualifying has finished
qualifying	`quali_gap_pct`	% slower than pole (best Q1/Q2/Q3 lap, clipped at 10 %)	Qualifying lap times

Group	Feature	Definition	Why it is valid before the race
qualifying	`grid_minus_quali`	Places lost or gained to penalties	Grid and qualifying, both pre-race
form	`drv_podium_rate_last5`	Driver's podium share in the previous 5 races	`shift(1)` excludes the current race
form	`drv_avg_finish_last5`	Average finishing position, previous 5 races	Same
form	`drv_finish_rate_last5`	Share of the previous 5 races finished	Same
form	`team_podium_rate_last5`	Team podiums per car, previous 5 races	Same, per constructor
form	`drv_circuit_podium_rate`	Driver's podium share in earlier visits to this circuit	Earlier visits only
standings	`drv_champ_pos_before`	Championship position before the race	Standings of the previous round
standings	`drv_points_share_before`	Driver points ÷ leader's points	Previous round
standings	`drv_wins_before`	Wins so far this season	Previous round
standings	`team_champ_pos_before`	Constructors' championship position	Previous round
standings	`team_points_share_before`	Team points ÷ leading team's points	Previous round
context	`career_starts`	Number of earlier F1 starts	Counts earlier races
context	`driver_age`	Age on race day	Date of birth and race date
context	`home_race`	Driver's country = circuit country	Static facts
context	`field_size`	Number of starters	Known after qualifying

Leakage control.

- The dataset stores standings **after** each race, so standings for race *k* are taken from round *k−1* of the same season. In round 1, points and wins are 0 and championship position is missing.
- Form features use `shift(1)` before the rolling window. An automated check recomputed the driver form feature for 300 random rows from strictly earlier races only, and all of them matched.
- Form features are computed on the full 1950–2024 history before filtering to 2003+, so early-2003 rows keep their pre-2003 history.
- **Excluded:** finishing position, points, laps, time, status, fastest lap, the current race's lap times and pit stops, and current-race standings.



Evidence (hypothesis-driven, on training years only). `quali_pos`, `grid_pos` and `quali_gap_pct` have the highest mutual information with the target, followed by team form and standings. The podium rate falls from 79 % (P1) to under 1 % (P15+), from 70 % (gap < 0.2 %) to about 0 % (gap > 2 %), and rises from 2 % to 67 % as team form goes

from 0 to above 0.7. Points *share* is used instead of raw points because the points system changed (10 points for a win in 2003, 25 from 2010). Context features have near-zero mutual information. They were kept as a hypothesis and tested by ablation (Section 7).

5. Split and preprocessing

We use a **temporal split by season**, with whole races kept inside one set:

Set	Seasons	Races	Rows	Podium rate
Train	2003–2018	300	6,383	14.1 %
Validation	2019–2021	60	1,198	15.0 %
Test	2022–2024	68	1,354	15.1 %

Why temporal. The model's real use is predicting future races from past ones. A random split would place later seasons in training, leaking future team strength and overstating performance. The test seasons also begin the 2022 regulation era, which makes them a demanding check of generalization. Hyperparameters, decision thresholds and model selection use the training and validation sets only; the test set is evaluated once.

Preprocessing (fitted on the training set only, inside pipelines):

- **Logistic regression and neural network:** median imputation plus missing-indicator columns, then standardization (mean 0, standard deviation 1). Both models are sensitive to feature scale.
- **Random forest:** median imputation only, because trees are scale-invariant.
- **Gradient boosting:** native missing-value handling, no preprocessing.

Preprocessing changes the range of the features but not their shape: missing values drop to 0 % and the imputed values form a spike at the training median.

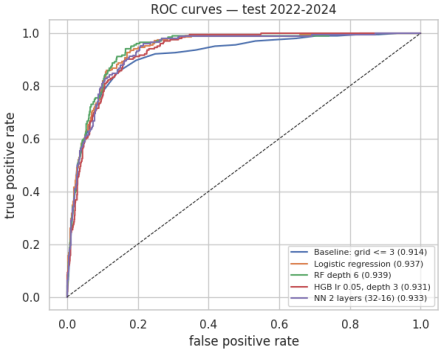
6. Models and results

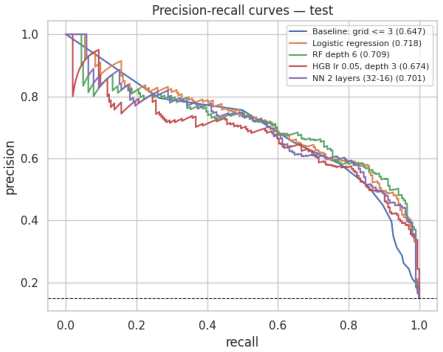
Metrics. We report ROC-AUC, PR-AUC, F1, precision and recall on the training, validation and test sets. **PR-AUC is the primary metric**: podiums are rare (about 15 %), the useful output is a ranking of likely podium finishers, and ROC-AUC is inflated by the many easy negatives (0.914 even for the grid rule). **F1 is the secondary metric**: it scores the actual yes/no decision, at a threshold chosen to maximize F1 on validation (0.33 for the selected model).

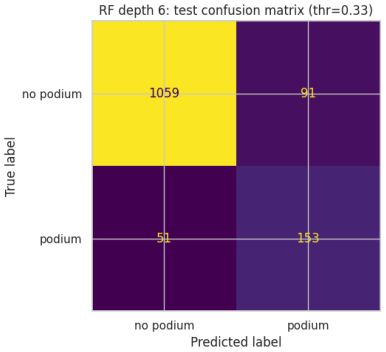
Models (nine learned variants plus a baseline):

- **Baseline:** the grid rule (top 3 on the grid = podium).
- **Logistic regression:** the statistical baseline model.
- **Random forest:** 300 trees, minimum leaf size 5, max depth 6, 10 or unlimited.
- **Histogram gradient boosting:** 300 iterations, (learning rate, depth) = (0.05, 3), (0.05, 6), (0.1, unlimited).
- **Shallow neural network:** scikit-learn `MLPClassifier`, ReLU, Adam, mini-batch 64, L2 penalty 1e-3, with 1 hidden layer (16) or 2 hidden layers (32–16). Training stops on validation PR-AUC with patience 15, restoring the best epoch. We used scikit-learn rather than TensorFlow so the notebook fits a 1 GB Kaggle session.

Model	Val PR-AUC	Val ROC-AUC	Val F1	Test PR-AUC	Test ROC-AUC	Test F1	Test precision	Test recall
Baseline: grid ⩽ 3	0.677	0.917	0.717	0.647	0.914	0.649	0.650	0.647
Logistic regression	0.735	0.938	0.729	**0.718**	0.937	0.672	0.550	0.863
RF depth 6 (selected)	**0.744**	**0.940**	**0.733**	0.709	**0.939**	**0.683**	0.627	0.750
RF depth 10	0.736	0.938	0.727	0.686	0.937	0.670	0.608	0.745
RF unlimited	0.737	0.939	0.726	0.684	0.936	0.671	0.621	0.730
HGB lr 0.05, depth 3	0.728	0.937	0.715	0.674	0.931	0.646	0.589	0.716
HGB lr 0.05, depth 6	0.698	0.929	0.704	0.640	0.922	0.661	0.607	0.725
HGB lr 0.1, no depth limit	0.669	0.924	0.703	0.626	0.920	0.647	0.601	0.701
NN 1 layer (16)	0.711	0.932	0.726	0.707	0.932	0.657	0.576	0.765
NN 2 layers (32–16)	0.726	0.933	0.728	0.701	0.933	0.681	0.596	0.794







Discussion.

- Every learned model beats the grid baseline on PR-AUC.

- The **random forest with depth 6** has the best validation PR-AUC and F1, so it is selected. On test it finds 153 of 204 podium finishers (recall 0.75) with 91 false alarms (precision 0.63).
- Logistic regression is marginally higher on test PR-AUC (0.718 vs 0.709). This is within noise for 1,354 rows, and the choice cannot be revised after seeing test results.
- Unconstrained trees overfit clearly: unlimited-depth boosting reaches training PR-AUC 1.000 but only 0.669 on validation.
- The shallow networks match logistic regression but not the forest, which is typical for about 6,400 rows of tabular data.

Limitations of the models:

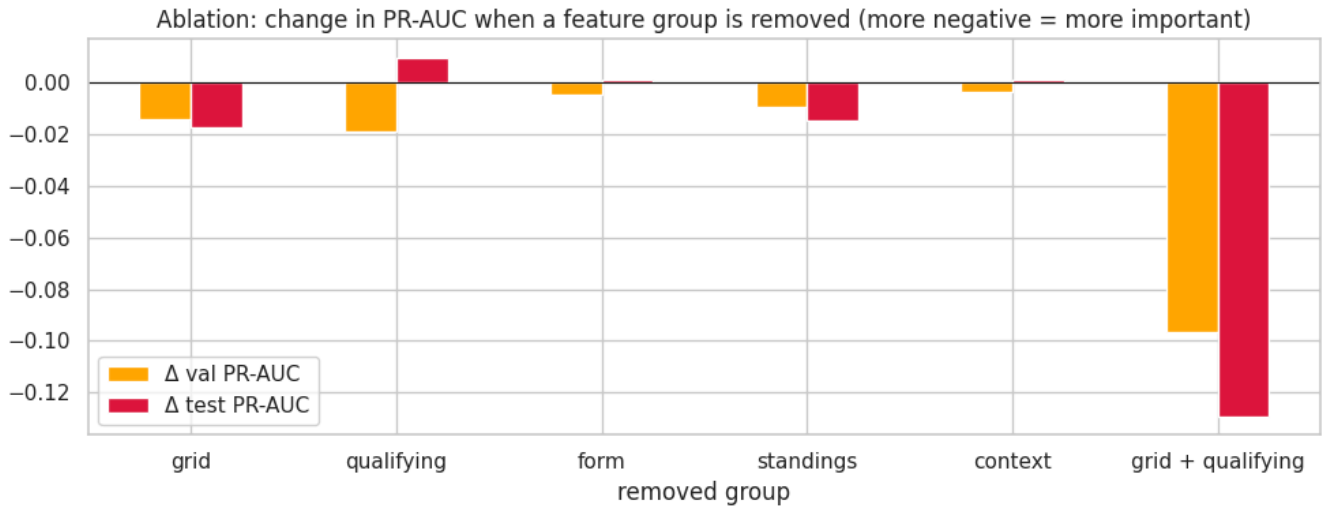
- Logistic regression captures only linear effects.
- Random forests give poorly calibrated probabilities and are opaque.
- Gradient boosting overfits without depth limits.
- The neural network needs scaled, imputed inputs and gains nothing over simpler models at this data size.

7. Experiments

7.1 Feature-group ablation

We removed one feature group at a time and retrained the selected model (random forest, depth 6).

Removed	Val PR-AUC	Test PR-AUC	Δ val	Δ test
nothing (all 20 features)	0.744	0.709	—	—
grid	0.730	0.692	−0.014	−0.018
qualifying	0.726	0.719	−0.019	+0.010
form	0.740	0.710	−0.005	+0.001
standings	0.735	0.694	−0.009	−0.015
context	0.741	0.710	−0.004	+0.001
grid + qualifying	**0.648**	**0.580**	**−0.097**	**−0.129**
only grid kept	0.677	0.647	−0.067	−0.062



Findings:

- Removing any single group costs at most about 0.02 PR-AUC, because the groups overlap. Grid and qualifying carry nearly the same information unless a penalty applies, and standings and form both describe car strength.
- Removing **grid and qualifying together** costs 0.097 (validation) and 0.129 (test), the largest drop: starting position is the core signal.
- Standings give the most consistent single-group contribution.
- Context and form add little on top of the other groups. Changes of ±0.01 on test are within noise.
- The grid group alone matches the baseline (0.677), so the other groups together add about 0.07 PR-AUC.

7.2 Preprocessing-order experiments

Each experiment changes only the order or scope of one step (logistic regression, the model most sensitive to preprocessing).

Experiment	Correct order (val PR-AUC)	Alternative (val PR-AUC)	Effect on the data
E1: imputer and scaler fitted on train only vs on all seasons	0.7347	0.7345 (leaky)	Test-year statistics leak into training
E2: impute → scale vs scale → impute	0.7347	0.7318	`drv_circuit_podium_rate` mean/std 0.02/1.01 vs −0.14/0.92

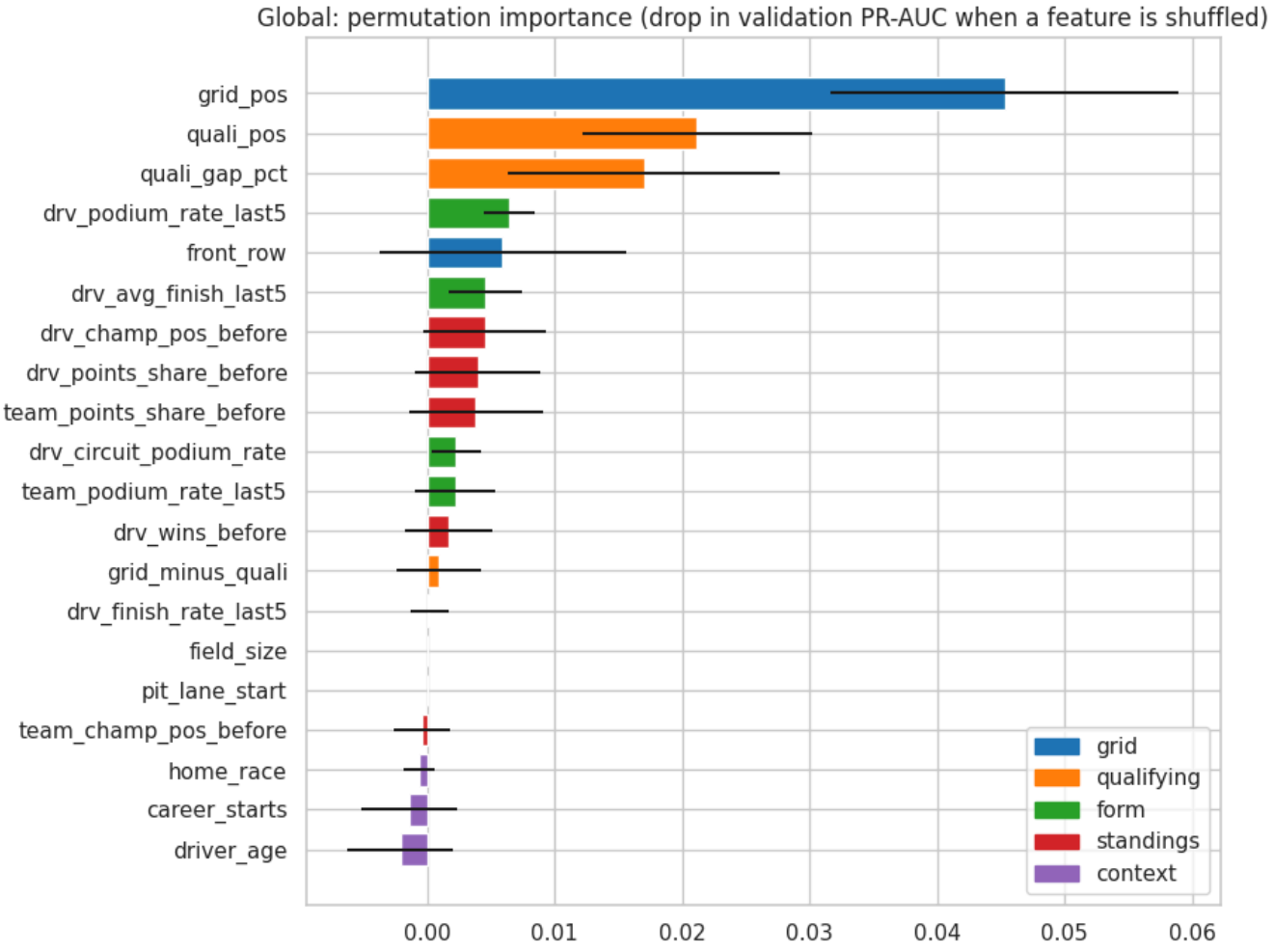
Experiment	Correct order (val PR-AUC)	Alternative (val PR-AUC)	Effect on the data
E3: form computed on full history, then filtered vs filtered to 2003+ first	0.7347	0.7343	Missing 2003 form values 82 → 412; mean `career_starts` in 2003: 79.9 → 7.1

All three change validation PR-AUC by less than 0.003, so the pipeline is robust. Each alternative order is either leaky (E1) or distorts feature values (E2, E3), which justifies the chosen order on correctness grounds.

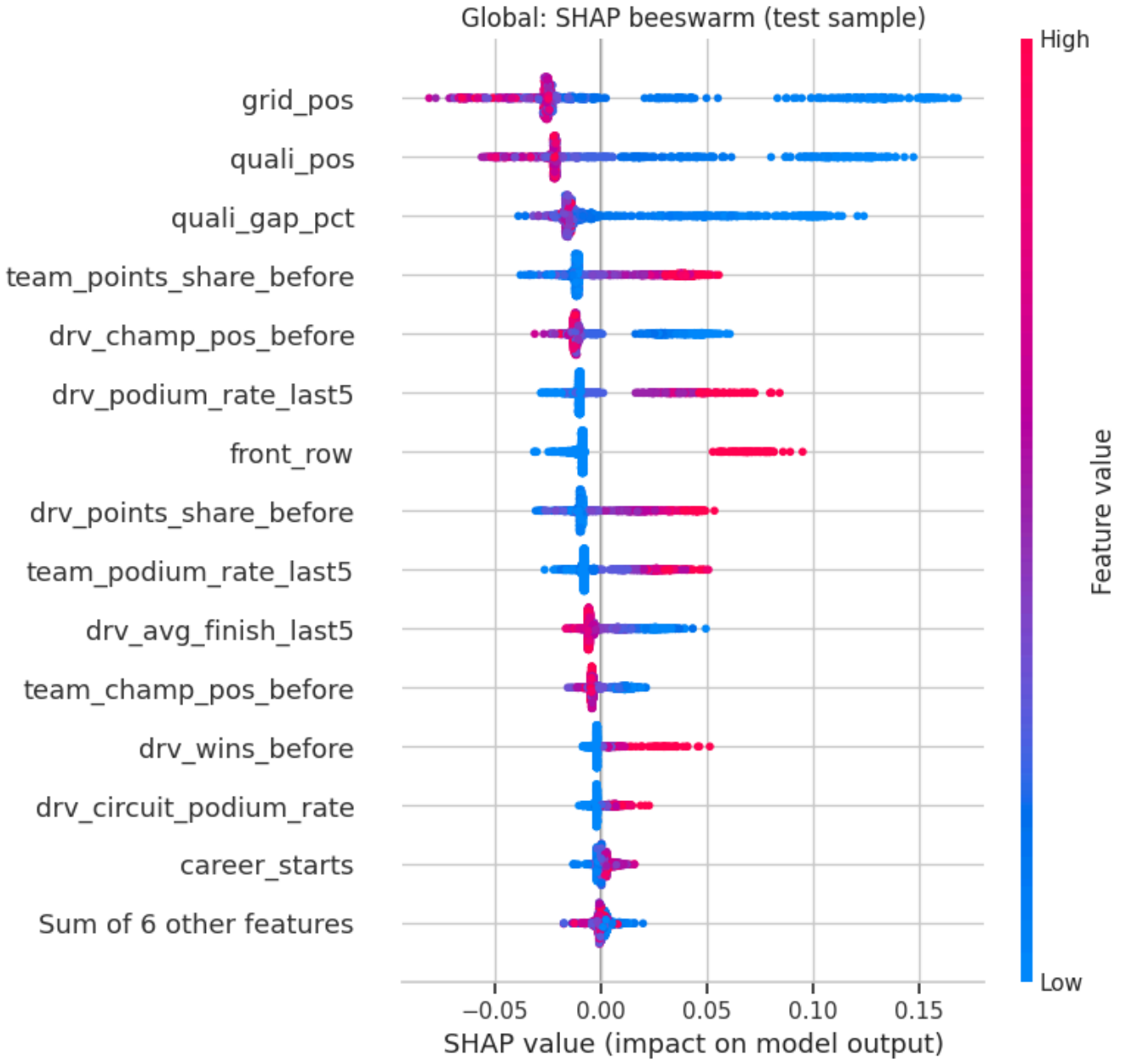
8. Explainability

> **Explanations are not causal effects.** A feature being important means the model *relies on it to make predictions*. It is not evidence that the feature *causes* podium finishes. For example, a high team points share does not cause a podium; it signals a fast car, which causes both good grid positions and podiums.

8.1 Global explanations



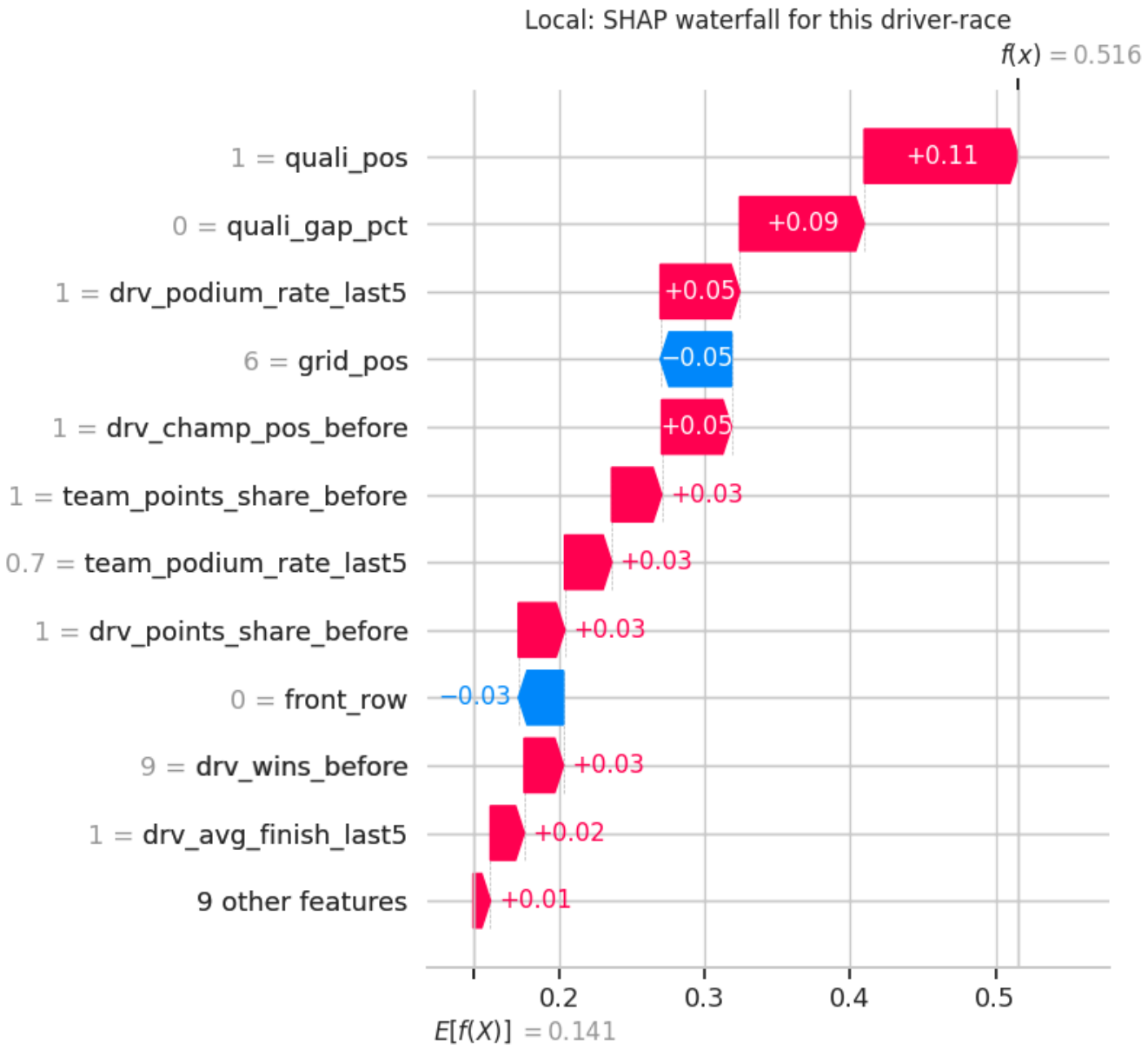
Permutation importance (validation set, PR-AUC, 10 repeats). Shuffling `grid_pos` lowers PR-AUC by about 0.045, `quali_pos` and `quali_gap_pct` by about 0.02 each, and form and standings features by about 0.005 each. Context features have importance near zero or below, so they are noise to the model.



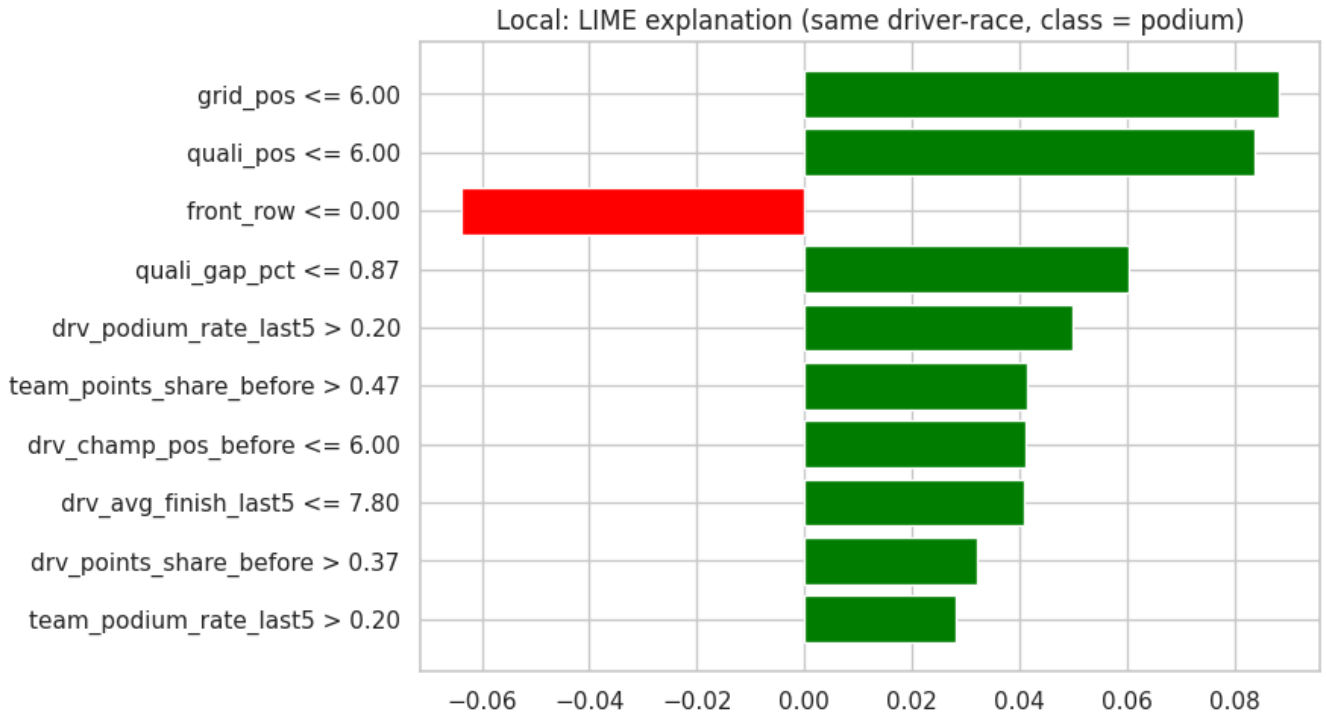
SHAP (TreeExplainer). The same features rank highest. A low grid or qualifying position (blue = near the front) pushes predictions toward a podium by up to +0.15. So do a small gap to pole, a high team points share, a strong recent podium rate and `front_row = 1`. Permutation importance and SHAP agree with the ablation in Section 7.

8.2 Local explanation: Max Verstappen, 2023 Belgian Grand Prix

Verstappen set the fastest qualifying time, took a 5-place gearbox penalty, and started 6th. The model predicted **P(podium) = 0.52**, above the 0.33 threshold, so it predicted a podium. He won.



SHAP waterfall. From the average prediction E[f(X)] = 0.141, `quali_pos = 1` (+0.11) and `quali_gap_pct = 0` (+0.09) push most strongly upward. Recent form, championship position and team strength add +0.03 to +0.05 each. `grid_pos = 6` (−0.05) and `front_row = 0` (−0.03) pull down. The final prediction is f(x) = 0.516.



LIME. The conditions `grid_pos <= 6` and `quali_pos <= 6` (+0.08 each), `quali_gap_pct <= 0.87` (+0.06) and the form and standings conditions support a podium; `front_row <= 0` (−0.064) is the only clear push against.

Interpretation. Both methods tell the same story: the grid alone (6th) suggests a low podium chance, but qualifying pace reveals the fastest car. This is the case the qualifying feature group was designed to capture.

9. Inference function

`predict_podium(record)` takes a dictionary of pre-race features for one driver in one race. It returns the podium probability, the yes/no decision at the validation-chosen threshold (0.33), the list of missing features, and any unrecognized keys. Missing fields are filled by the model pipeline's training-median imputation.

Input	Output
Complete record: Sergio Pérez, 2022 Saudi Arabian GP (pole)	P(podium) = 0.603, predicted podium, no missing features (actual: 4th)
Same record without the 8 qualifying and standings fields, plus an unknown key `typo_feature`	P(podium) = 0.251, no podium; reports the 8 missing features and flags `typo_feature` as ignored

With fields missing, the probability moves toward the average because imputation assumes a typical driver for those fields. Returning `missing_features` makes this transparent to the user.

10. Limitations and conclusion

Known limitations:

- **Qualifying-coverage drift.** There is no qualifying data before 1994 and coverage is patchy until 2002. The model covers 2003–2024 only and does not transfer to earlier eras.
- **Era sensitivity.** Regulation changes (2014 hybrid, 2022 ground effect) reshuffle the competitive order. The test seasons come from a new era, and form features need several races to adapt after a rule change.
- **Missing context.** There is no weather, tyre strategy, car upgrade or safety-car information.
- **Scoring changes.** Points systems changed in 2003, 2010 and 2019; points share reduces but does not remove this effect.
- **Small test set.** 68 races and 204 podiums, so differences of about ±0.01 PR-AUC are within noise.
- **Q2 and Q3 assumptions.** Nationality-to-country mapping was done by hand, and ambiguous retirement statuses are excluded, so the mechanical rates are lower bounds.
- **Explanations describe model reliance, not causation.**

Conclusion. After auditing and cleaning 14 relational tables, we built a leakage-safe table with one row per driver per race (2003–2024, 20 features). A depth-limited random forest predicts podium finishes on unseen 2022–2024 races with PR-AUC 0.709, compared with 0.647 for the starting-grid rule. Starting position (grid or qualifying) is the core signal; season strength and recent form add the rest, while context features add nothing measurable. Ablation, permutation importance and SHAP all agree on this. The data-engineering analysis shows that front-row conversion rankings shift between eras (ρ = 0.45), that there is no home advantage once grid position is controlled for (odds ratio 0.92, 95 % CI 0.78–1.08), and that mechanical retirements fell from 9.2 % to 6.0 % of starts between 2014–2021 and 2022–2024.